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# An Ensemble Version of the E-OBS Temperature and Precipitation Data Sets

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**Key Points:**
- An improved uncertainty estimate is provided through the generation of multiple realizations
- The new data set is broadly consistent with the original version
- The largest differences occur in the precipitation grids

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**Abstract** We describe the construction of a new version of the Europe-wide E-OBS temperature (daily minimum, mean, and maximum values) and precipitation data set. This version provides an improved estimation of interpolation uncertainty through the calculation of a 100-member ensemble of realizations of each daily field. The data set covers the period back to 1950 and provides gridded fields at a spacing of 0.25° × 0.25° in regular latitude/longitude coordinates. As with the original E-OBS data set, the ensemble version is based on the station series collated as part of the ECA&D initiative. Station density varies significantly over the domain, and over time, and a reliable estimation of interpolation uncertainty in the gridded fields is therefore important for users of the data set. The uncertainty quantified by the ensemble data set is more realistic than the uncertainty estimates in the original version, although uncertainty is still underestimated in data-sparse regions. The new data set is compared against the earlier version of E-OBS and against regional gridded data sets produced by a selection of National Meteorological Services. In terms of both climatological averages and extreme values, the new version of E-OBS is broadly comparable to the earlier version. Nonetheless, users will notice differences between the two E-OBS versions, especially for precipitation, which arises from the different gridding method used.

## 1. Introduction

Gridded data sets formed from the interpolation of station-derived meteorological observations remain a principal source of information for monitoring the climate system. While global, monthly, and often anomaly-based data sets provide information about large-scale forcing of the climate, data sets that resolve synoptic-scale-to-mesoscale processes through the provision of daily values at a relatively fine spatial resolution are vital for many applications, especially those involving extremes. Such data sets require a high density of stations—especially when gridding precipitation—and are generally only produced for individual countries due to restrictions concerning the full sharing of daily meteorological observations. Nonetheless, several initiatives have constructed high-resolution data sets for larger regions through the cooperation of National Meteorological Services (NMSs) across neighboring countries (e.g., Isotta et al., 2014; for the European Alpine region). However, high spatial resolution daily gridded data that cover continental regions in a consistent manner are required by many users and it is in this context that the E-OBS data set was developed for Europe (Haylock et al., 2008).

E-OBS originally consisted of daily mean, minimum, and maximum, temperature values and daily precipitation totals gridded at a resolution of approximately 25 km and was developed to provide validation for the suite of Europe-wide climate model simulations produced as part of the EU ENSEMBLES project (Hofstra et al., 2008). In later work mean sea level pressure was added as a gridded variable (van den Besselaar et al., 2011). While E-OBS remains an important data set for model validation (Lenderink, 2010; Min et al., 2013; Nikulin et al., 2011), it is also used more generally for monitoring the climate across Europe (Lavaysse et al., 2017; van der Schrier et al., 2013; van Oldenborgh et al., 2016), particularly with regard to the assessment of the magnitude and frequency of daily extremes.

In recent years several regional reanalysis data sets have become available that provide high-resolution, sub-daily data sets for the European domain. These reanalyses have drawn upon the experience gained in the generation of global-scale reanalyses and have retained many of the desirable properties of their global

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counterparts through the provision of a range of meteorological variables in a consistent manner, for both the surface and at intervals throughout the vertical atmospheric profile. For Europe significant progress has been made in the development of regional reanalysis data sets in two European Union seventh Framework Programme (EU FP7) projects: *European Reanalyses and Observations for Monitoring* and *Uncertainties in Ensembles of Regional Re-Analyses* (Jermey & Renshaw, 2016). In contrast to global-scale reanalyses, regional reanalyses are able to resolve meteorological processes at a much finer scale and solve many of the limitations of station-based gridding exercises with respect to providing spatial and temporal generalizations of climate fields. However, while regional reanalysis data sets have shown great potential, the computing resources required to produce such data sets are considerable, and this has meant that the reanalyses can, in practice, only currently generate data for the last ~20 years. Furthermore, regional reanalyses still show large systematic errors (Bach et al., 2016; Dahlgren et al., 2016; Isotta et al., 2015), particularly for precipitation since rain-gauge data are not currently assimilated into reanalysis data sets. Hence, Europe-wide data sets, such as E-OBS, that are relatively quick to produce and which stretch back over longer time periods remain important tools for climate monitoring and model validation. Indeed, much of the work described in this paper was carried out as part of the Uncertainties in Ensembles of Regional Re-Analyses project, with the aim of providing a more consistent data set against which the reanalysis data sets constructed during the project could be evaluated. This evaluation role of E-OBS will remain important for the ERA5 global reanalysis, which is currently in production and will ultimately provide a high-resolution (31 km) data set back to 1950.

In this paper we describe and evaluate a new interpolation method for the E-OBS temperature and precipitation data set. A particular focus in this study is the provision of a better estimate of uncertainty of the gridded data through the generation of an ensemble of daily realizations. Few gridded climate data sets provide estimates of uncertainty, with even fewer estimating this value from an ensemble of realizations. Notable exceptions are the works of Clark et al. (2006) and Newman et al. (2015, who created ensemble data sets for a region of western Colorado and for the conterminous United States, respectively), the global HadCRUT4 data set (Morice et al., 2012), and the gridded data set for Finland constructed by Aalto et al. (2016). In addition to the more rigorous estimation of uncertainty afforded by such ensemble data sets (Beguería et al., 2016), they also allow uncertainty estimates to be propagated through derived products, such as extremes indices.

The interpolation method described in this paper marks a major departure from the method used in the current operational version of E-OBS, as documented by Haylock et al. (2008). In the following section we describe the station data used in the data set, and particularly how the density of stations has changed with successive versions of E-OBS. The technique used to calculate the ensemble is also described in that section. In section 3 the quality of the interpolation is assessed against a selection of withheld station data, against the current operational version of E-OBS (v16.0) and against a selection of national/supranational gridded data sets produced by NMSs across Europe. Throughout this paper the term *v16.0e* is used to denote the ensemble data set, while *v16.0* refers to the current operational version. The variables maximum, minimum, and mean daily temperature are referred to as *tx*, *tn*, and *tg*, respectively; *rr* is used to denote the daily total of precipitation variable.

## 2. The Station Data and Gridding Method

### 2.1. The Underlying Station Data

The station data collated by the ECA&D initiative (Klein Tank et al., 2002; Klok & Klein Tank, 2009) form the basis of E-OBS. These data are supplied by many NMSs and other providers across Europe and the Middle East, although due to restrictions concerning the exchange of data, the number of stations available to E-OBS is generally lower than the total number potentially available, or the number that is used in most national/supranational gridded data sets. Although most station series are quality controlled by the respective agencies, the series are subjected to a further quality control procedure following incorporation into ECA&D. These data are then blended with neighboring series to form more temporally complete series (The ECA&D Team, 2012; van der Schrier et al., 2013), and these blended series are used in E-OBS. To update the series to near real time, data are used from synoptic messages (SYNOP) distributed via the Global Telecommunications System. The SYNOP data are replaced when validated data become available from the NMSs, but these validated series are often delayed due to the data being subjected to additional quality control procedures by the respective agencies (van den Besselaar et al., 2012). A range of different methods are used to calculate the daily values, and this is often dependent upon the convention used by a given data provider. For example, daily mean temperatures may be the average of maximum and minimum temperatures, or may be the

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average of hourly readings, or another frequency such as three hourly. This variation is not accounted for in the gridding of the data. Furthermore, uncertainties in the measurement of the variables are not taken into account in the gridding, or in the ensemble-based uncertainty estimates, which only quantify interpolation uncertainty. While such errors are likely to be present in both the temperature and precipitation data, measurement errors in precipitation may be substantial, particularly at high-elevation or high-latitude stations (Legates & Willmott, 1990; Neff, 1977; Yang et al., 1999a).

Since the initial construction of E-OBS by Haylock et al. (2008), many more station series have been added to ECA&D (Figure 1a), with an increase from approximately 1,200 to approximately 3,700 stations for temperature, and from approximately 2,500 to approximately 9000 stations in the case of precipitation. However, since only certain agencies have increased the number of stations in ECA&D, there has been an increasing disparity in station density across the domain, with relatively many stations across central Europe and Scandinavia, and many fewer toward the south and east of the domain. It should also be noted that many more stations are available in ECA&D east of 50°E, but since these are not incorporated into E-OBS, they are not plotted in Figure 1a. In addition to the highly variable station density, the number of stations available for gridding also varies significantly over time with many fewer stations before circa 1961 and a decrease in density after circa 2000; this has been a deficiency in E-OBS since its inception (Figure 1 b, cf. Figure 2 in Haylock et al., 2008). The network of stations shown in Figure 1 for v16.0 are also used in this paper for the new ensemble data set (v16.0e). Hence, comparisons in this paper between the data sets only relate to the different gridding methods used.

### 2.2. The Gridding Method

In the original version of E-OBS (Haylock et al., 2008) the gridding procedure consisted of the following five stages:

1. Daily proportions or anomalies were calculated at each station relative to the station monthly total (precipitation) or mean (temperature).
2. The monthly values were interpolated to a high-resolution grid (0.1° rotated-pole grid) using a trivariate thin-plate spline.
3. The daily proportions or anomalies were gridded to the same high-resolution grid using ordinary kriging.
4. The gridded daily values were multiplied or added to the respective gridded monthly totals or means to form the daily absolute values.
5. The high-resolution gridded data were averaged to a coarser grid resolution to provide grid box spatial averages.

In E-OBS v16.0e we adopt a two-stage process to produce the daily fields: (1) the daily values are initially fitted with a deterministic model, to capture the long-range spatial trend in the data, and (2) the residuals from this model are then interpolated using a stochastic technique (Gaussian Random Field, GRF, simulation) to produce the daily ensemble. Monthly values are also used in the interpolation, since the relationship of altitude to the meteorological fields can be difficult to discern in daily resolution data, particularly for precipitation (Masson & Frei, 2014). However, in E-OBS v16.0e these values are incorporated as a covariate in the deterministic model. As in E-OBS v16.0, each day is gridded independently, and in the case of temperature independent of the other temperature elements.

#### 2.2.1. The Spatial Trend Model Used for the Temperature Variables

The spatial trend in the daily temperature variables (tn, tg, and tx) is captured by fitting a Generalized Additive Model (GAM) to the station values:

$$y = f_1(lon, lat, alt) + f_2(bg) + \epsilon$$

where the daily temperature data ($y$) are modeled as a smoothed function of longitude (*lon*), latitude (*lat*), and altitude (*alt*) using a reduced-rank thin-plate spline, plus a smoothed function of the monthly mean, background field values of temperature (*bg*) using a cubic spline (Wood, 2003, 2006). GAMs are an extension of generalized linear models, which extend simple linear regression by allowing the independent variable to take a distribution other than Gaussian. GAMs extend this further by allowing the use of nonlinear functions between the response variable and the independent variables (Hastie & Tibshirani, 1990). In the models used in this paper we only make use of the smoothing feature of the GAMs and assume that the dependent data (transformed in the case of precipitation) are normally distributed.

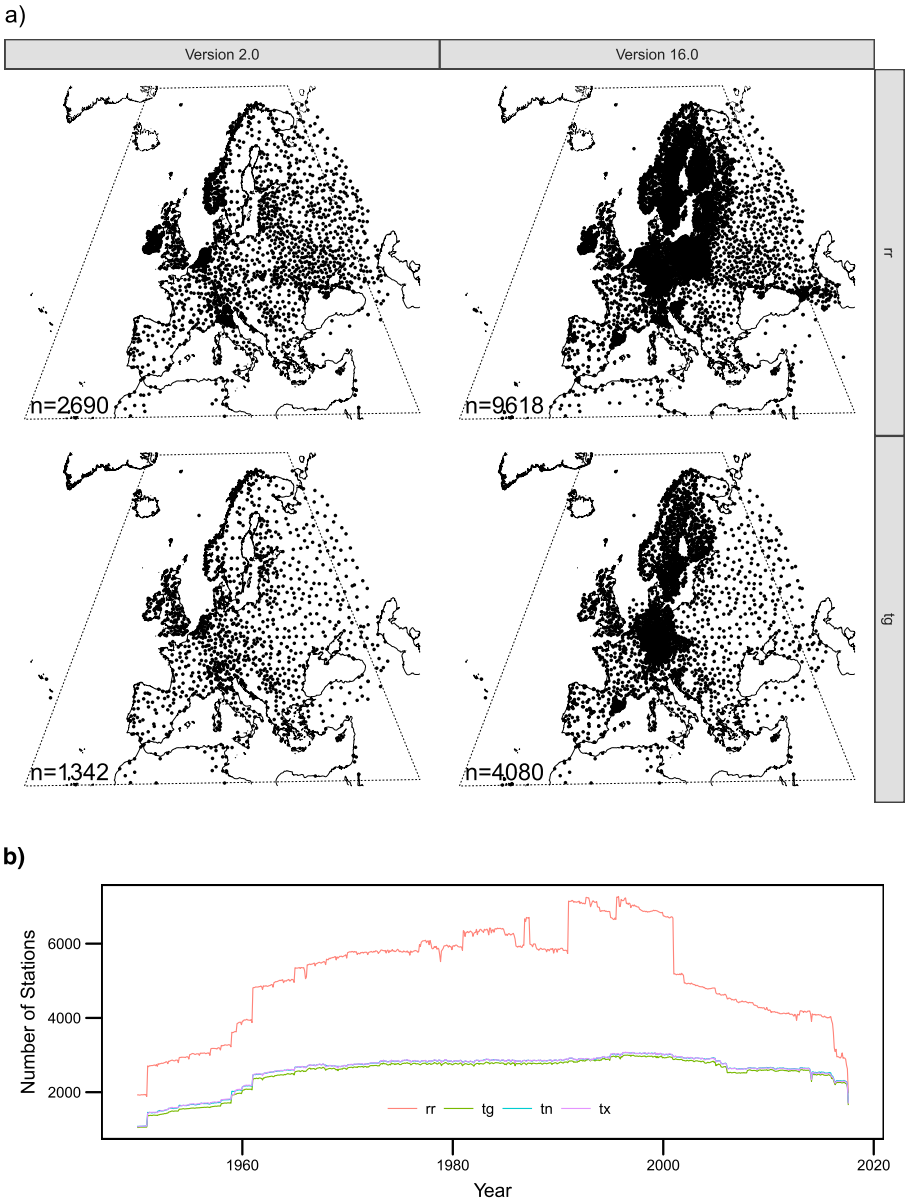



**Figure 1.** The location of stations used in the gridding of mean temperature and precipitation in versions 2.0 (released August 2009) and 16.0 (released September 2017) of E-OBS (a). These are stations that are used at least once in the gridding and may not be present for the full duration of the gridded data set. The dotted line indicates the gridding area used in the ensemble version of E-OBS (in the Lambert Azimuthal Equal-Area projection, EPSG:3035). The plot in (b) shows the numbers of stations for each month from January 1950 to August 2017.

The GAM is fitted using penalized likelihood maximization using the penalized iterative least squares method (Wood, 2006). An optimal fitting of the functions ($f_1$ and $f_2$) is achieved by minimizing the Generalized Cross Validation statistic, which is equivalent to a leave-one-out cross validation (Hastie & Tibshirani, 1990) and is used to select an optimal smoothing parameter ($\lambda_1$, $\lambda_2$) in each of the functions $f_1$ and $f_2$, respectively. The values of $\lambda_1$ and $\lambda_2$ in the penalized regression models used here range between 0 and 1, with 0 indicating an exact fit to the data and 1 indicating a least squares regression fit. An eigen decomposition is used to achieve the rank reduction in the functions $f_1$ and $f_2$, which requires the prior selection of the parameter $k$ to provide an upper limit of $k-1$ to the effective degrees of freedom (EDF) of the model; the actual EDF is still selected using Generalized Cross Validation, and the parameter $\lambda$ retains its control on the flexibility of the spline under this limit (Wood, 2006). Since the function $f_1$ is used in this analysis to remove the long-range spatial trend from the data, we restrict the EDF (i.e., the inverse of $\lambda$) of the function by setting a deliberately low basis dimension



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($k = 50$) relative the number of stations ($n > 1,000$). Since $k$ is set to a low number relative to the true spatial variation in temperature, the model residual ($\epsilon$) contains significant local scale spatial autocorrelation, which is modeled using GRFs (e.g., Schabenberger & Gotway, 2005; see section 2.3). In the function $f_2$, a value of $k = 10$ is chosen using the heuristic tests described by Wood (2006, 2014).

The background field (*bg*) is used in these models to simplify the field for establishing the fitting of the trivariate spline. Tests with and without *bg* indicated improved model fit through its inclusion as a model variable. While the background field includes an altitude element, the addition of station altitude is still necessary in $f_1$. This captures the day-to-day variation in the environmental lapse rate, and its incorporation as a third variable in the function $f_1$ generates a spatially varying lapse rate (Hutchinson, 1995a; Hutchinson et al., 2009). The original version of E-OBS also incorporated a daily lapse rate, but since the variable was used as the drift element in the kriging of the daily anomaly values, the lapse rate was fixed across the domain.

The monthly background field values (*bg*) used in the GAM are produced using a full trivariate spline in the same way as the original E-OBS data set (cf. the reduced-rank approximations used here for the daily fields). As with E-OBS v16.0, and following the work of Hutchinson (1995b), altitude was scaled in the unit of kilometers, with latitude and longitude scaled in degrees longitude/latitude; this scaling was also used in the daily model ($f_1$). Interpolated monthly values, rather than actual station monthly means, are used in the daily model so that stations without complete monthly values could be used in the daily model. This also allows the use of more monthly values — which are generally more widely available than daily data — in the formation of *bg* in future updates. As in the earlier versions of E-OBS, station-based altitude values are used for model fitting, while GTOPO30 digital elevation model (DEM) values of altitude are used for the interpolation. Although GTOPO30 has been superseded by GMTED2010 (Danielson & Gesch, 2011), we continue to use GTOPO30 in this analysis to ensure against the introduction of a confounding effect resulting from the use of differing elevation data. We note, however, that the use of the different DEMs has a significant effect on the interpolation and needs to be investigated further. In earlier versions of E-OBS, the monthly thin-plate spline suffered from overfitting, and in most cases an exact interpolator was produced, that is, with zero smoothing and a spline that fitted exactly through each data point. This is often a feature of such splines and results in a model that performs poorly in data-sparse regions and is vulnerable to outliers in the station data (Hutchinson, 1998a). To guard against this in E-OBS v16.0e, the degrees of freedom of the thin-plate spline were inflated by a factor of 1.1. To prevent overfitting of spline models, Kim and Gu (2004) suggested applying an inflation factor of 1.4, but this produced fields that were too smooth in this case. The value of 1.1 was determined by comparing the interpolation against the monthly fields used in E-OBS v16.0.

#### 2.2.2. The Spatial Trend Model Used for Precipitation

To calculate the long-range spatial trend for precipitation, a model similar to the temperature GAM was used; however, this was applied to nonzero precipitation values ($y$):

$$\sqrt{y} = f_1(lon, lat) + f_2(\sqrt{bg}) + \epsilon$$

In contrast to the spatial trend model for the temperature models, altitude was not directly incorporated into the daily precipitation models, as this did not significantly improve the fitting of the model. This corresponds to assertions made by Masson and Frei (2014) that the precipitation-altitude relationship is often difficult to ascertain at the daily resolution since the variability is dominated by synoptic-scale weather systems. To remove some of the skewness in the data, the precipitation values ($y$) and monthly background precipitation totals (*bg*) were square root transformed prior to fitting (Hutchinson et al., 2009). This also ensures that all interpolated values, when converted back to the unit of millimeters, are nonnegative. In this model $k = 100$ is used in the function $f_1$.

The *occurrence* of precipitation (as a binary field where rr >0 mm was set to 1) was gridded separately from *accumulations* using a full thin-plate spline. This was then used to mask the daily fields. Precipitation was deemed to have occurred at a given grid box where the gridded value exceeded 0.5, after Hutchinson et al. (2009). The gridding of precipitation accumulation and occurrence separately follows the method used in earlier versions of E-OBS and reduces the inflation of the numbers of wet days. This inflation is a common problem in interpolations of daily precipitation and occurs as a result of the different spatial scales of precipitation accumulations and precipitation occurrence (Hutchinson et al., 2009) and ultimately arises from the zero-bounding nature of precipitation (Hewitson & Crane, 2005). The use of the thin-plate spline method for determining the occurrence of precipitation in the gridded fields produces a result that is similar to the



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indicator-kriging method used in the original version of E-OBS. To prevent overfitting of this spline model, an EDF inflation factor of 1.4 was used after Kim and Gu (2004).

### 2.3. Generating the Ensemble

In the original version of E-OBS a measure of uncertainty for each daily field was provided. These values were calculated by summing a climatological standard error value, calculated over a fixed base period (1961–1990, after the method described by Hutchinson and Tingbao (2013)), and the daily kriging uncertainty (derived using the technique developed by Yamamoto, 2000). Climatological standard error values were used instead of the monthly standard error, which would match the monthly background field interpolation, due to the significant computation time required for this calculation. The possibility of providing a better estimate of uncertainty in the interpolated field through the generation of an ensemble of grids for each day was explored in the initial construction of E-OBS (Haylock et al., 2008; Hofstra et al., 2009). This was rejected, however, due to the significant computational burden that the procedure entails.

In E-OBS v16.0e uncertainty is estimated using stochastic simulation to produce an ensemble of realizations of each daily field. A set of 100 spatially correlated GRFs are generated for each day that are conditional on the residuals ($\epsilon$) from the deterministic spatial trend model described above. The spatial structure of the random fields is defined through the calculation of an empirical variogram up to a maximum lag distance of 1,000 km for temperature and 900 km for precipitation. A distance-decay model for each day was then calculated by fitting an exponential variogram model to the distance classes:

$$c = c_0 + c(1 - e^{-h/a})$$

The nugget ($c_0$) was fixed at 30% of the precision of the variables to take into consideration measurement precision (Aalto et al., 2016; i.e., 0.03 for both temperature, °C, and precipitation, mm), while the sill ($c$) and range ($a$) parameters were determined using a weighting that placed a higher weight on distance lags with more observations. This weighting was also used in the variogram model fitting in the original version of E-OBS (Haylock et al., 2008). Prior to fitting of the variogram and generation of the random fields, the coordinates of the data had been converted to a Lambert Equal-area projection.

In the use of a single variogram across the domain an assumption is made that the correlation structure is only dependent on the spatial lag, and not the location. This assumption is often made in such applications (e.g., Haylock et al., 2008; Newman et al., 2015) but can be an oversimplification for continental-scale data as the spectral characteristics of the true fields can vary considerably across the domain, particularly for precipitation. This assumption of stationarity may also be unrealistic in the present application on account of the varying characteristics of the spatial trend captured by the GAM, which results from the large variations in station coverage across the domain; this is assessed in section 3.2. Testing was carried out using different variograms across the domain; this, however, introduced artifacts in the gridded data, and particularly in the ensemble spread.

The GRFs were produced on a uniform grid at approximately 12-km regular grid spacing, which is equivalent to the grid spacing of the master grid used in earlier versions of E-OBS. The spatial trend values were interpolated to the same grid, and these values were added to each of the ensemble fields. The precipitation values were then squared to convert the units of the interpolated field to millimeters. For the temperature variables, grid cells with fewer than four stations in a distance of 500 km were set to missing. For precipitation the cutoff distance was set to 450 km. These distances are similarly used in E-OBS v16.0, but in v16.0e the values represent half the distance used in the calculation of the empirical variograms; in the case of E-OBS v16.0 these values mark the maximum lag distance. To produce *best guess* fields, the mean across the 100 ensemble members is calculated. The ensemble spread (90% confidence range) was calculated as the difference between the 95th and 5th percentiles calculated from the 100 members at each grid box.

Since the GRF simulations are conditioned on point (station) values, the simulation produces values that are representative of point values (Hewitson & Crane, 2005). In E-OBS the aim is to produce values that represent area-averaged values. This is achieved by aggregating the fields from 0.1° to the 0.2° resolution. These fields were then interpolated to the E-OBS v16.0 grid using bilinear interpolation. This method was preferred to simply averaging all values from the 0.1° grid that fall within the 0.25° final grid as that method produced spurious grid patterns in the precipitation fields that resulted from discrepancies in the number of grid points that fell within each final grid box.



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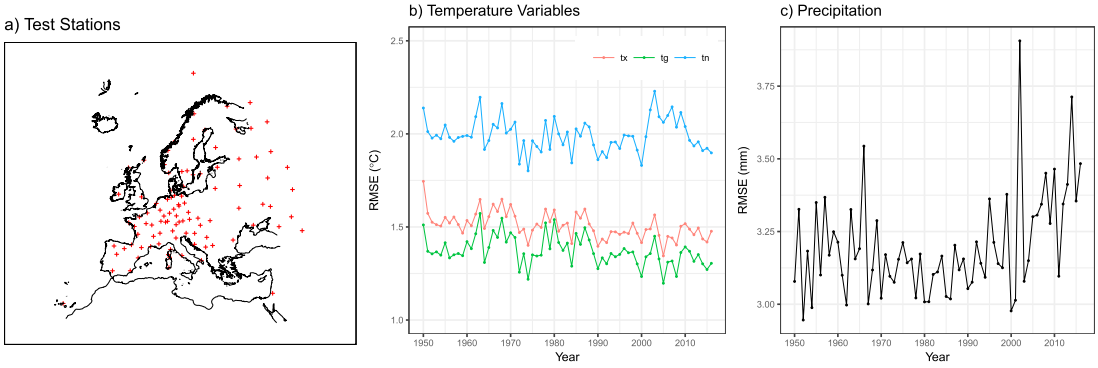



**Figure 2.** The locations of the 100 withheld test stations (a) and the yearly root-mean-square error (RMSE) values between interpolated values and recorded values at those stations for the temperature variables (b) and precipitation (c).

In the production of the precipitation data set, the fixed precipitation occurrence mask for the respective day (section 2.2.2) is used to mask all of the ensemble members. This method was chosen as it reduces the overall computing time for the production of the gridded fields and produces consistency across the ensemble but means that the ensemble does not provide an estimate of the likelihood of precipitation occurrence and hence neglects the uncertainty in occurrence of small-scale precipitation events.

A particular challenge in the production of E-OBS is the spatiotemporal variation in station coverage (see section 2.1), and this is particularly the case in producing a reliable measure of uncertainty through the use of GRFs. The GAMs are used to satisfy the assumption of spatial stationarity in the GRFs. However, with the variation in station coverage being large, a question arises as to the relative proportion of spatial variance that is captured by the GAM compared to that captured by the GRF. As described in section 2.2.1, the EDF of the $f_1$ function are deliberately restricted so that only the long-range spatial trend is captured by the model. There is a risk, however, with the variation in station coverage that the GAM will explain more short-scale variance in the field in areas of high station density, compared to regions of sparse coverage. As a consequence the ensemble spread could be overestimated (underestimated) in areas of dense (sparse) station coverage. The value of $k$ was chosen in the function $f_1$ to be low enough to only capture the large-scale spatial trend. This is more successful for temperature than for precipitation, as in the case of precipitation only nonzero values are interpolated, which results in large variations in sampling. Two possible improvements to the gridding method could be made to alleviate this problem: (1) a fixed set of stations could be used that are complete over the entire 1950–2016 period or (2) the spatial trend and residual simulation could be jointly estimated. The first option would result in a very small sample of available stations, and the gridding would be largely confined to central northern European areas. The embedding of the ensemble simulation into the GAM would provide a more feasible course of action, but such models are notoriously computationally demanding. Nonetheless, this could be investigated for future updates of E-OBS.

## 3. Results and Discussion

### 3.1. Assessment Against Withheld Station Data

To assess the quality of the interpolation, we have withheld 100 stations from the sample of stations available for gridding, and we have then compared interpolated values against the recorded station values at those locations. Reference stations were selected initially by only retaining stations that were more than 90% complete over the 1950–2016 period for all four variables. A spatially even coverage of 100 stations was then selected using minimax sampling, which selects stations from the total number of available stations such that the maximum distance from the station to any of the other stations is minimized (Figure 2a). Since the density of stations varies significantly over the domain (Figure 1), the use of a fixed sample of stations ensures that spatial bias in the testing is reduced and also ensures that the reference station values take no part in either the background (monthly) field interpolation in the first stage of gridding or the daily interpolations. A full leave-one-out cross validation was not possible given the number of stations involved and the time required to complete the interpolation.

We have calculated error statistics (root-mean-square error, RMSE, and mean absolute error) between the interpolated values and reference stations over the period 1950–2016 (Table 1). In this analysis the



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**Table 1**
*Seasonal and Annual RMSE and MAE Statistics Between the 100 Reference Station Values and Interpolated Values*

| Variable | Season | RMSE | MAE |
|---|---|---|---|
| rr | Winter | 2.80 | 0.99 |
| | Spring | 2.74 | 0.95 |
| | Summer | 3.63 | 1.25 |
| | Autumn | 3.52 | 1.16 |
| | Annual | 3.20 | 1.09 |
| tg | Winter | 1.78 | 0.93 |
| | Spring | 1.25 | 0.76 |
| | Summer | 1.15 | 0.74 |
| | Autumn | 1.24 | 0.76 |
| | Annual | 1.37 | 0.80 |
| tn | Winter | 2.41 | 1.35 |
| | Spring | 1.94 | 1.19 |
| | Summer | 1.67 | 1.12 |
| | Autumn | 1.89 | 1.21 |
| | Annual | 2.00 | 1.22 |
| tx | Winter | 1.71 | 0.96 |
| | Spring | 1.44 | 0.92 |
| | Summer | 1.53 | 0.98 |
| | Autumn | 1.33 | 0.84 |
| | Annual | 1.51 | 0.92 |

*Note.* The statistics are calculated over the period 1950–2016 and are in the units of degrees Celsius for the temperature variables and millimeters for precipitation. RMSE = root-mean-square error; MAE = mean absolute error.

interpolated values are the mean across the 100 realizations. For the temperature variables, RMSE values range from 1.15 °C to 2.41 °C. This error, however, is highly dependent upon the temperature variable or season considered. The highest errors are observed for daily minimum temperature and during the winter season. The error values also vary seasonally in the case of daily precipitation totals, with RMSE values reaching a minimum of 2.74 mm in the spring, and a maximum of 3.63 mm in the summer. This is a reflection of the fact that winter precipitation will be more associated with fronts, and summer precipitation is more likely to be convective in nature.

The RMSE values also vary over time, although in the case of the temperature variables the stratification in RMSE values remains constant over the time period (Figure 2). In the case of precipitation the RMSE values exhibit relatively large year-to-year variation. The results for the tx and tg variables display a gradual reduction in RMSE values since 1950, although there is an indication of a step to slightly lower error values after 1990, especially in the results for tx. Interestingly, in the case of tn there is a step to higher RMSE values after the year 2001. An examination of the annual RMSE values on a station-by-station basis (not shown) reveals that these step changes only occur in some of the reference stations. Hence, the temporal changes in RMSE are likely attributable to inhomogeneities in the reference stations rather than any particular change in the gridding. In light of these results, we reiterate the message of previous studies (Cornes & Jones, 2013; Hofstra et al., 2009) that caution should be exercised when using the E-OBS data set for the examination of long-term trends across Europe as the station data have not undergone any homogeneity *correction* at present. Homogeneity *testing* is carried out on the stations (The ECA&D Team, 2012), but this information is not taken into account when gridding the data. Inhomogeneities in the gridded data are also likely as a result of the marked change in station numbers over time.

The discrepancy in error between tx and tn highlights a further feature of the E-OBS data that must be taken into consideration by data users: since

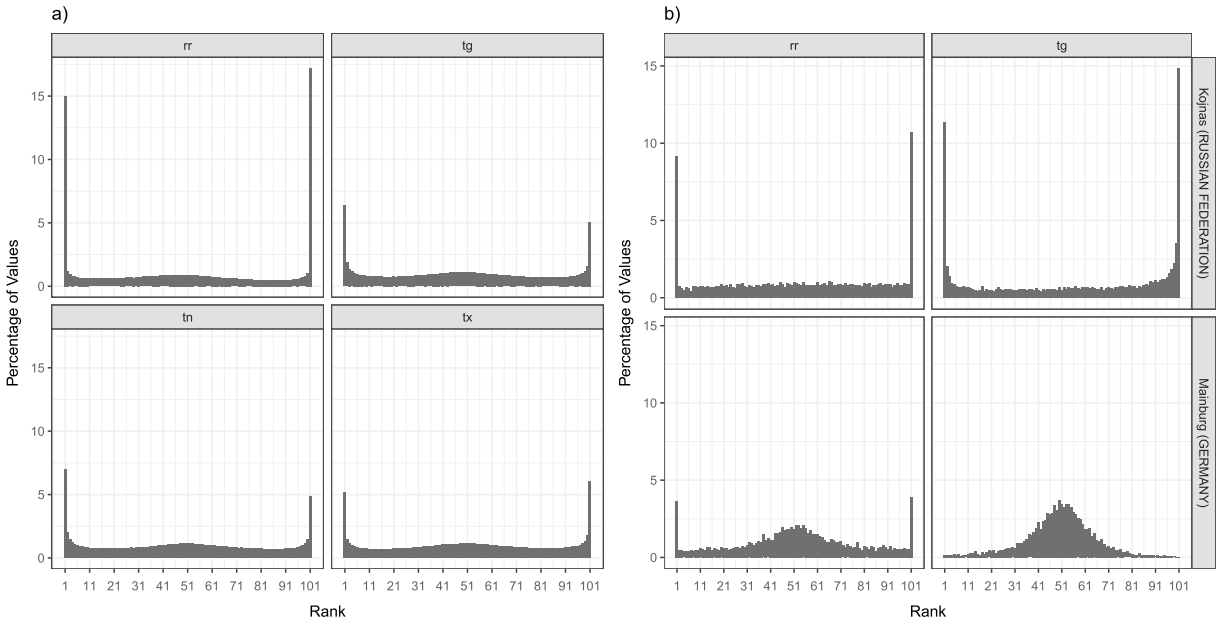



**Figure 3.** Rank histograms calculated for the E-OBS v16.0e ensemble compared to the 100 withheld stations, across all stations over the period 1971–2000 (a) and for two selected stations (b) over that period.



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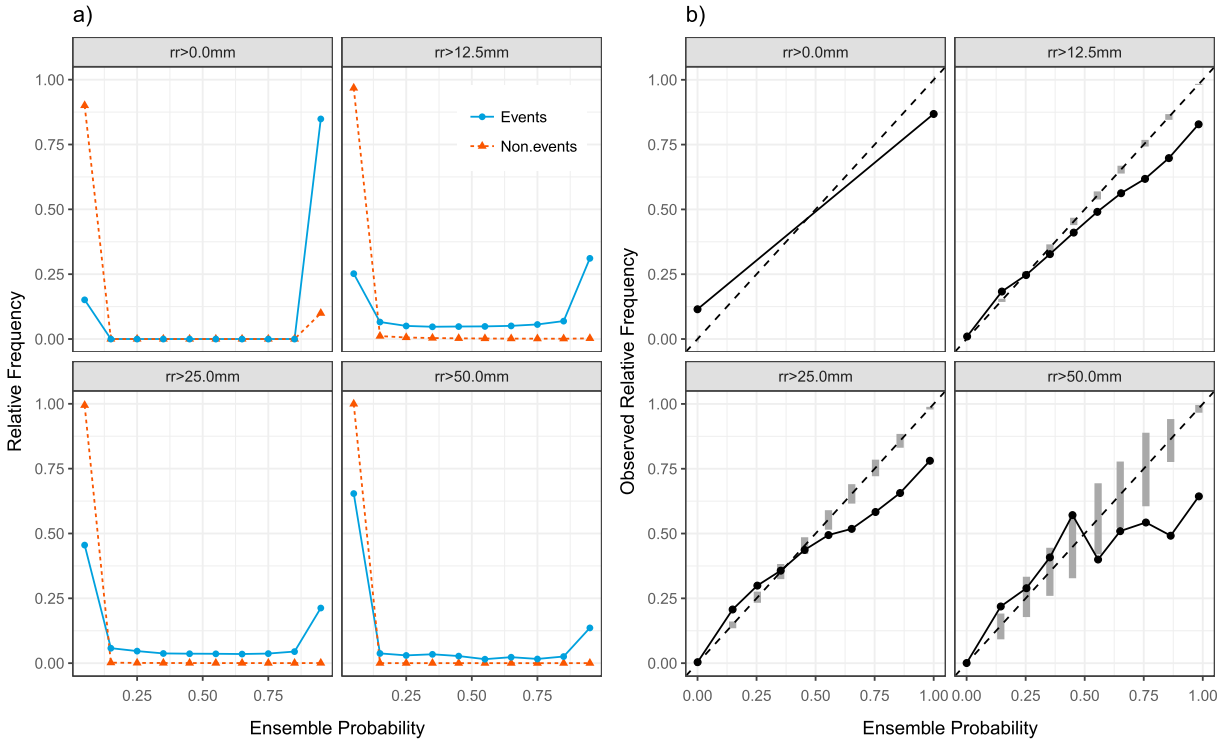



**Figure 4.** Discrimination diagrams (a) and reliability diagrams (b) for four observed precipitation intensities. The gray vertical bars in (b) indicate the 5–95% uncertainty range calculated after the method described in Bröcker et al. (2007).

tx and tn are gridded independently, there is no guarantee that individual daily tx values are greater than tn, despite the input station data being checked for this property. Similarly, there is no guarantee that interpolated tg values will be the mean of the tx and tn interpolations at the respective grid boxes, as may be expected given that this is one of the more widely used method of calculating tg in the station data. The number of occurrences of tx < tn varies over the time period, with up to 0.4% of grid cells experiencing the problem in the last 5 years of the series, which seems to be a result of the use of more Global Telecommunications System-derived data during that period; prior to that the average of yearly counts of tx < tn occurrences is 0.06%.

### 3.2. Features of the Ensemble

Rank histograms are a simple way of determining the reliability of an ensemble of realizations relative to observations (Wilks, 2006). They are calculated by ranking an ensemble from lowest to highest to form a series of $n + 1$ bins, given $n$ ensemble realizations; the highest and lowest bins are open ended. The corresponding observed value is then allocated to the appropriate bin. This procedure is then repeated for different observation sites and/or different times to generate a large sample. Using the withheld station observations from section 3.1, rank histograms have been calculated for the E-OBS ensemble (Figure 3). In the case of precipitation, values are only used when nonzero precipitation occurred in the ensemble members and observations, after Bach et al. (2016).

The rank histograms calculated for all of the 100 station sites over the period 1971–2000 (Figure 3a) show a distinctive U shape, with a much higher proportion of the observations falling in the first and last histogram bins compared to the central range of bins. This indicates that overall the ensemble is too optimistic in representing the range of uncertainty, that is, the ensemble samples are taken from a distribution with a lack of variability compared to the observations (Hamill, 2001). However, the nature of the histograms varies considerably depending on the location of the observation station. For stations in the regions of higher station density, such as Germany or Sweden (see Figure 1), the rank histograms typically show a higher relative frequency in the central bins, which indicates an overestimation of spread in the ensemble, that is, the estimates of uncertainty are overly pessimistic. Conversely, an underestimation of ensemble spread typically occurs in stations located in regions that are sparse in station coverage, such as Russia or Spain, despite the spread of



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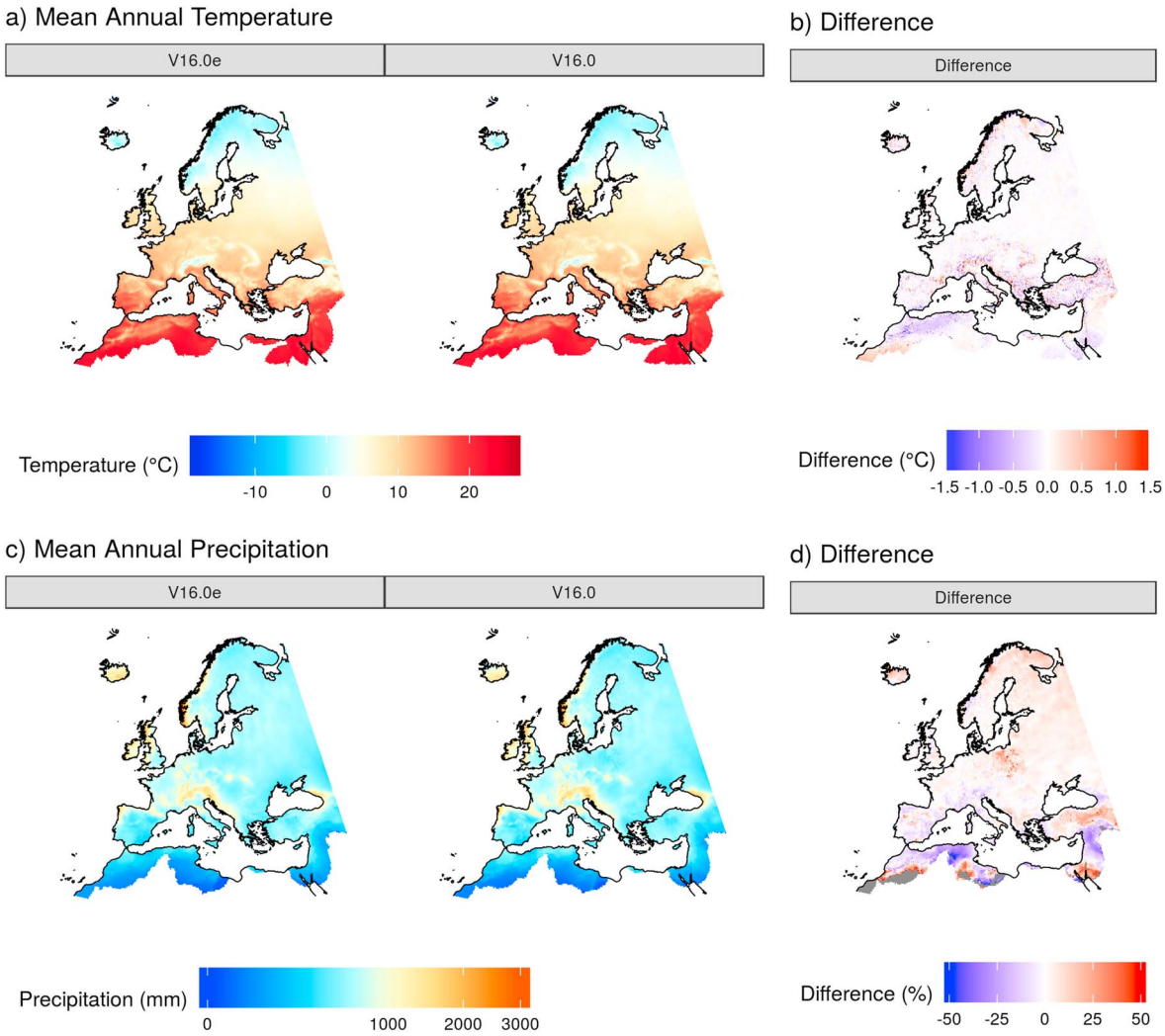



**Figure 5.** Climatology comparisons between E-OBS v16.0e and v16.0 for (a) mean annual temperature (tg) and (c) annual precipitation totals, calculated over the period 1981–2010. Indicated in (b) and (d) are the differences between the new and old E-OBS versions. In the case of precipitation (d) the difference is expressed as a proportion of mean precipitation totals from the old data set.

members being larger in such areas. Typical examples are shown in Figure 3b. This may result from the calculation of the ensemble using a single variogram for the entire region, which biases the spatial structure of the interpolated field, and hence the ensemble spread, to the spatial variation observed in the regions of higher station density. This feature may also result from a disparity in spatial variability captured in the GAM relative to the GRFs (see section 2.3). For precipitation the picture is more complicated, and even in areas of high station density an overdispersion of observations occur in the outer bins of the histogram, as demonstrated for the station *Mainburg* in Germany in Figure 3b. However, for that station a relatively high dispersion is also apparent in the central bins. This pattern may indicate different degrees of spread reliability under different precipitation regimes.

Discrimination and reliability diagrams are also useful ways of indicating the characteristics of an ensemble data set and are used here to provide further information about the v16.0e precipitation ensemble. Discrimination diagrams show the frequency with which a certain event (nonevent) probability is estimated by the ensemble in cases when the event has happened (not happened; Wilks, 2006). Following the example of Clark et al. (2006) and Newman et al. (2015) these diagrams have been calculated using four thresholds (Figure 4). The discrimination diagrams (Figure 4a) indicate that a good degree of separation exists in the discrimination of observed events in the lower precipitation categories, although there is an increasing overlap in the distributions for the higher categories. As a reflection of this, the absolute difference in means of the ensemble probabilities from the event/nonevent categories (the discrimination distance) is 0.52 for the 12.5-mm



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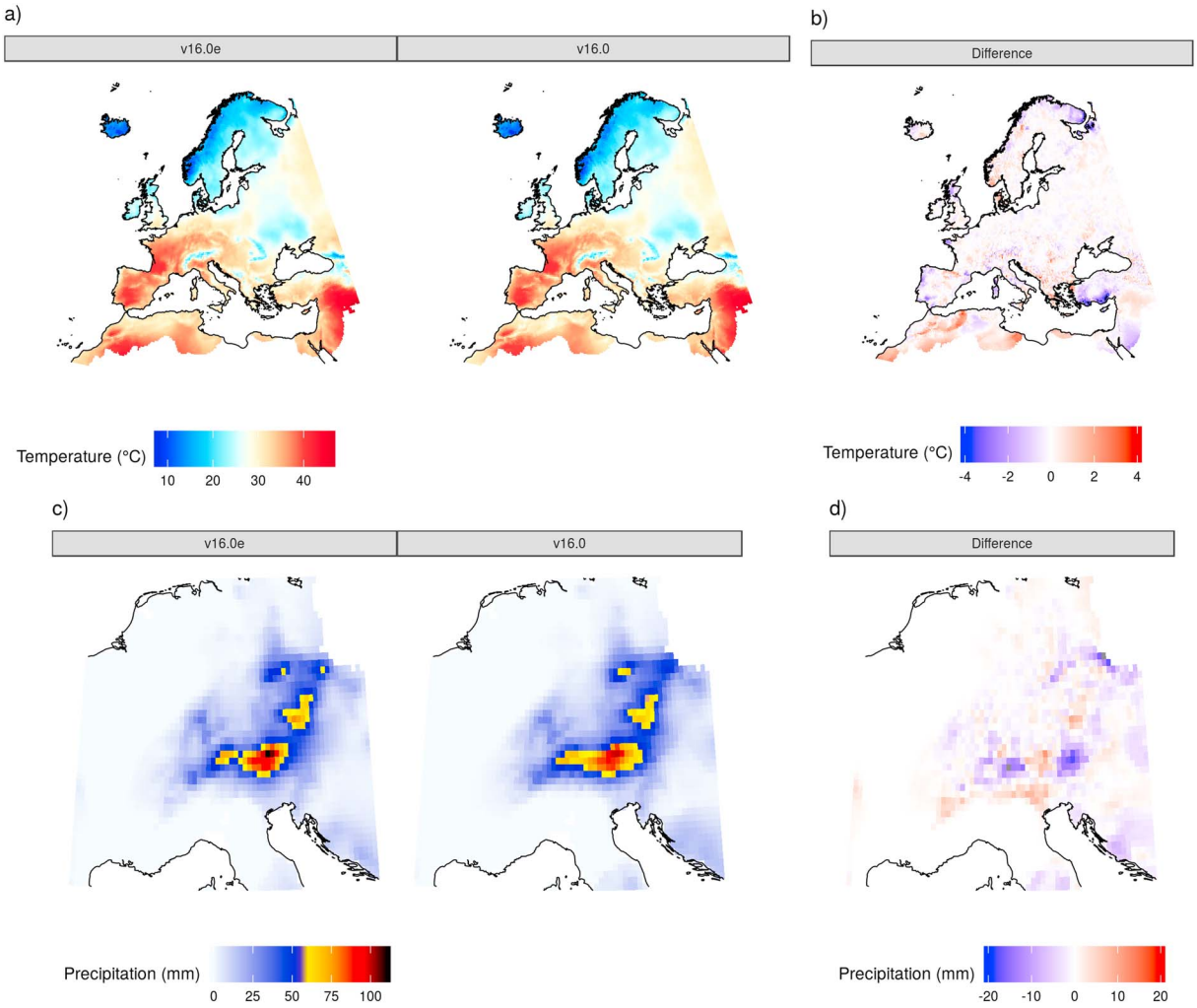



**Figure 6.** The gridded fields in E-OBS v16.0e and v16.0 for tx on 4 August 2003 (a) and rr on 1 June 2013 (c). The respective differences between the E-OBS versions are shown in (b) and (d).

threshold but reduces to 0.23 in the 50-mm threshold. It should be noted that the common masking of rainfall for all ensemble members on a given day results in a binary likelihood for $rr > 0$ mm in the ensemble probability; improvements could be made to the data set through the use of a probabilistic method to estimate precipitation occurrence.

In accordance with the results from the discrimination diagrams, the reliability diagrams (Figure 4b) indicate a slight dry bias in ensemble-estimated probabilities of precipitation occurrence below around 0.7 although these values are within the 5–95% consistency range. Above probabilities of around 0.6–0.7, a wet bias is apparent in the results, which is most apparent at the highest threshold level of $rr > 50.0$ mm.

In this section the ensemble values have been compared against station values. In contrast, the gridded data set aims to provide values that are representative of grid box averages. Hence, discrepancies would be expected in the comparisons, particularly for precipitation, and the results are consistent with interpolated values that smooth over station-scale values. The results for the temperature variables indicate, however, that the scale that is depicted in the interpolated values likely varies over the domain, with gridded values in areas with relatively high station density resolving the temperature field at a smaller scale than in data-sparse regions. A similar conclusion was reached by Hofstra et al. (2009) in their evaluation of the original version of E-OBS, but this discrepancy has likely become more pronounced as more stations have been added to certain areas (see Figure 1).



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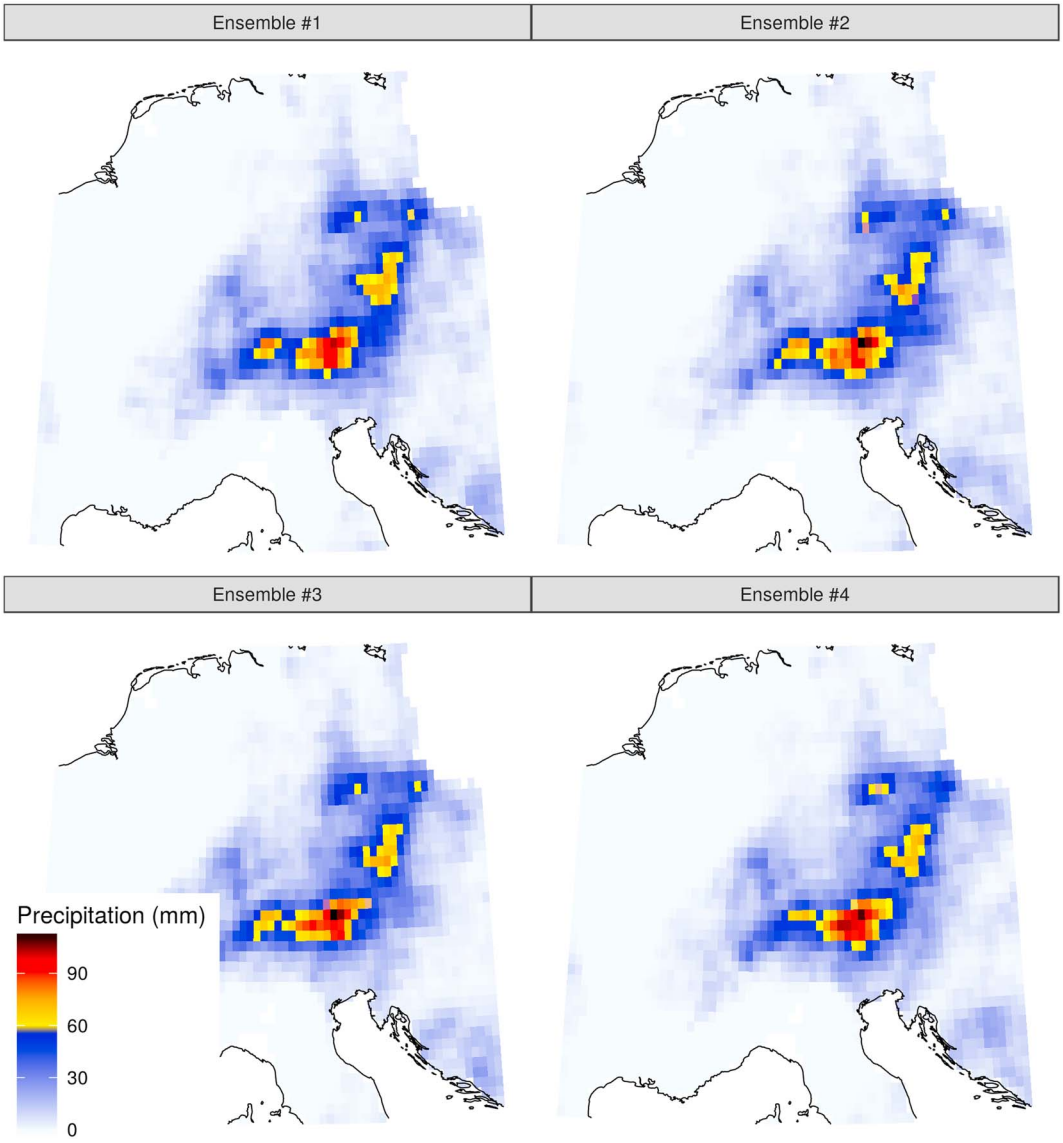



**Figure 7.** The first four ensemble members from E-OBS v16.0e for the precipitation event of 1 June 2013.

### 3.3. Comparison Against the Previous E-OBS Version

In Figure 5 we compare the climatological annual averages (totals in the case of precipitation) calculated over the period 1981–2010 from E-OBS v16.0 and v16.0e. In the case of temperature, the largest (negative) differences in the climatologies occur across North Africa and the Middle East. This is the region of lowest station density and hence these are the regions where the largest differences would be expected to occur. Across most other regions of Europe the differences are generally in the range ±0.5 °C, and across many areas approach 0 °C. Larger differences tend to occur across mountainous regions, particularly the Alps, and this is likely a result of the significantly different methods used to incorporate the environmental lapse rate in the two versions of E-OBS (as discussed in section 2.2.1).

In the case of precipitation (Figure 5c/5d) the largest differences also tend to occur across North Africa and the Middle East, which likely result from a combination of the low density of stations, the associated high degree of uncertainty in the interpolation, and the low-average rainfall totals recorded in these regions. However, v16.0e is also wetter than v16.0 by around 5–10%, particularly across eastern regions of the gridded region. This is most probably a result of the method used to scale the daily precipitation anomalies by monthly totals in v16.0, compared to the incorporation of altitude in the spatial trend model as a function of the monthly



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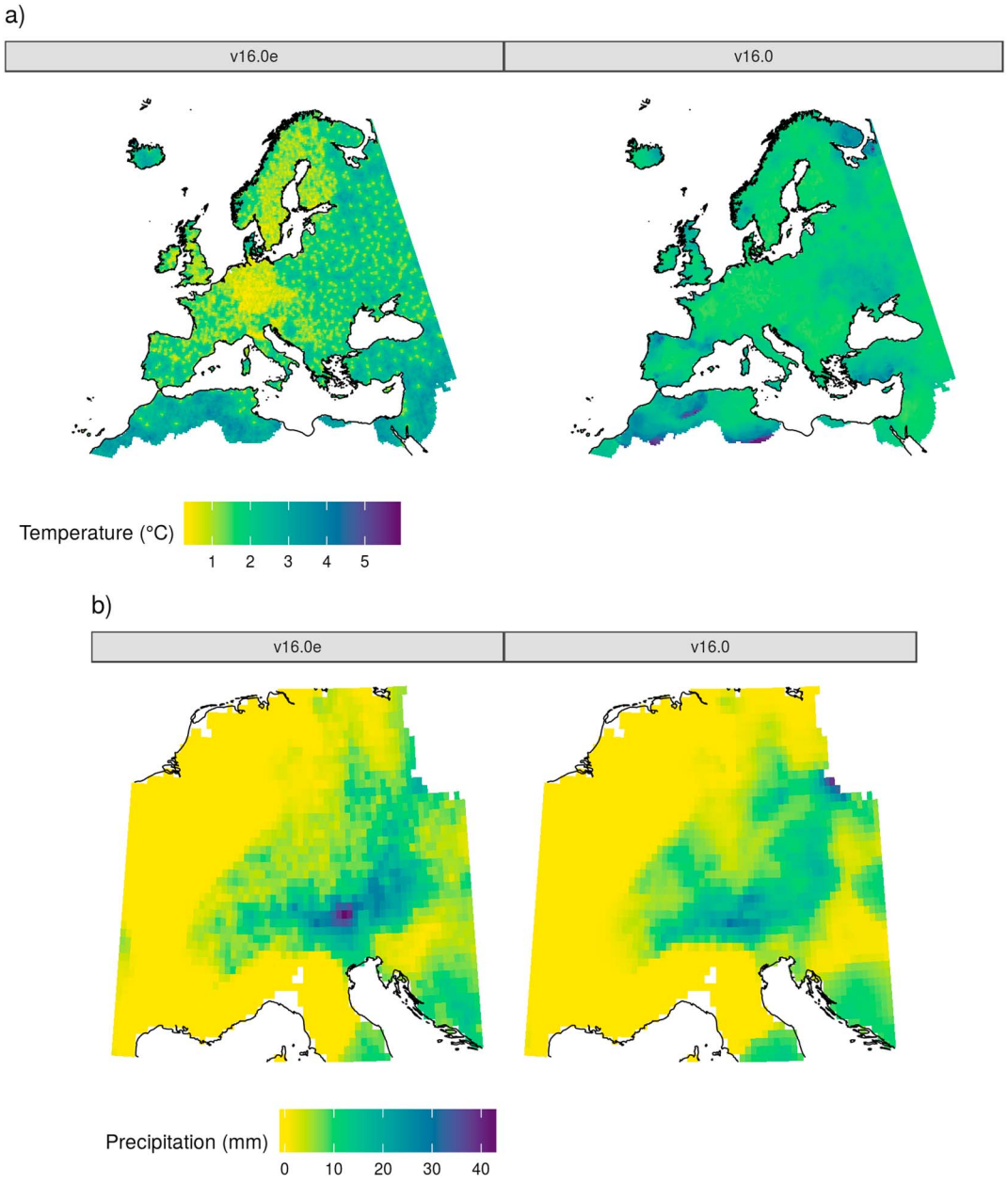



**Figure 8.** Uncertainty estimates in the tx gridded data for the 4 August 2003 event (a) and for the 1 June 2013 precipitation event (b).

interpolation in v16.0e, and demonstrates the sensitivity of precipitation grids to the choice of interpolation method (Herold et al., 2016; Sun et al., 2014).

To further illustrate the differences between the two versions of E-OBS, we have examined the interpolations of maximum daily temperature for 4 August 2003 and daily precipitation for 1 June 2013. Both of these events represent very extreme events in the ECA&D record: the 2003 event was the climax of the exceptionally hot summer of 2003 (García-Herrera et al., 2010), and the precipitation event of 2013 led to significant flooding across the Upper Danube catchment (Blöschl et al., 2013).

In the case of the 2003 event (Figure 6a), the two versions of E-OBS are broadly consistent, with differences being generally less than 1 ° C. As with the climatological mean values, the largest differences tend to occur across North Africa, and likely reflects the poor data coverage in this region. Differences of more than −4 °C



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**Table 2**
*Details About the NMSs Gridded Data Sets Used for Comparison Against E-OBS*

| Data set | Region | | | | | Version | Resolution | Reference |
|---|---|---|---|---|---|---|---|---|
| | | tn | tg | tx | rr | | | |
| CARPATCLIM | Carpathian basin | x | x | x | x | — | 10 km | Spinoni et al. (2015) |
| EURO4M_APGD | Greater Alpine | | | | x | 1.0 | 5 km | Isotta et al. (2014) |
| MeteoSwiss | Switzerland | x | x | x | | 1.2 | 1 km | Frei (2014) |
| MET_NO | Norway | | x | | x | 1.1 | 1 km | Mohr (2009) |
| PORT02 | Portugal | | | | x | — | ~20 km | Belo-Pereira et al. (2011) |
| SAFRAN | France | | x | | x | — | 8 km | Vidal et al. (2010) |
| SPAIN02 | Spain | x | | x | x | AA-3D v4 | ~10 km | Herrera et al. (2016) |
| UKCP09 | United Kingdom | | x | | | — | 5 km | Perry et al. (2009) |

*Note.* The resolution refers to the native resolution of the data set.

are also evident in certain other regions, notably southern Turkey and Western Russia, which also seem to be a result of the response of the different gridding methods to interpolating across data-sparse regions.

The spatial structure of the precipitation event of 2013 is broadly similar between the two versions of E-OBS, although differences of about 10 mm are apparent for certain grid cells (Figure 6 c). The most intense precipitation is more concentrated in v16.0e, whereas the region experiencing >75 mm is larger in v16.0. As with the precipitation climatologies described above, this is likely a result of the method used in v16.0 of scaling the daily gridded field to the monthly precipitation totals. It should be noted, however, that there is considerable uncertainty in the precipitation totals for this event in E-OBS since the highest precipitation totals occurred in a region with relatively sparse data coverage as well as being in a mountainous region. This uncertainty is demonstrated in Figure 7, which plots the first four ensemble members for this precipitation event.

In Figure 8 the uncertainty ranges for the two interpolations are plotted. In v16.0 this uncertainty is defined as 1.64 times the standard error (90% uncertainty, under the assumption of a normal distribution), whereas in v16.0e this is more strictly defined as the 90% range across the ensemble (see section 2.3). Despite this difference in uncertainty definition, in v16.0e these values are more closely related to station coverage than v16.0, which shows a much more smoothed field of uncertainty. This difference in spread between the two E-OBS versions is most pronounced for temperature, where the spread is much reduced at grid cells close to stations, and in data-rich regions, but is much larger in areas of low station density. In the rank histograms examined above, the ensemble was shown to be generally overly optimistic, but this appears to be a more realistic measure of uncertainty than was provided in E-OBS v16.0. The spread of uncertainty for precipitation is also much larger in data-sparse regions and is more closely linked to station density in v16.0e. However, the uncertainty for precipitation must also scale with interpolated totals (Hutchinson, 1995b, 1998b). Hence, in Figure 8b the relationship to station coverage is not as immediately apparent as in the temperature interpolation.

### 3.4. Comparison Against NMSs Gridded Data Sets

To further examine the differences between E-OBS v16.0e and v16.0, we have evaluated the data sets against several high-resolution regional data sets produced by various NMSs across Europe (Table 2). A variety of different interpolation procedures are used to construct these data sets, but they are generally developed using many more station data than are available to E-OBS, although this disparity varies by data set. In addition, since these data sets are developed for discrete regions of Europe, they suffer less from the constraints in E-OBS of the need to generalize the meteorological field across a large diverse domain, and with a greatly varying spatial density of station data. In the case of E-OBS, spatial variability is generalized using smoothing parameters fixed across the region, both in the GAM-derived spatial trend and in the simulated residuals (cf. the piecemeal method used by Lussana et al., 2017). For these reasons the NMSs data sets would be expected to provide a closer estimate of the true areal averages for their respective domains than E-OBS (Hofstra et al., 2008). The NMSs data sets were regridded to the E-OBS 0.25 ° × 0.25 ° regular grid using box averaging. The NMSs data are generally produced on a much higher resolution grid than E-OBS (Table 2), and hence, these aggregated values are expected to be comparable to the box average values of E-OBS. However, in the case of the PORT02 the resolution of 20 km is comparable to E-OBS. In that case nearest neighbor



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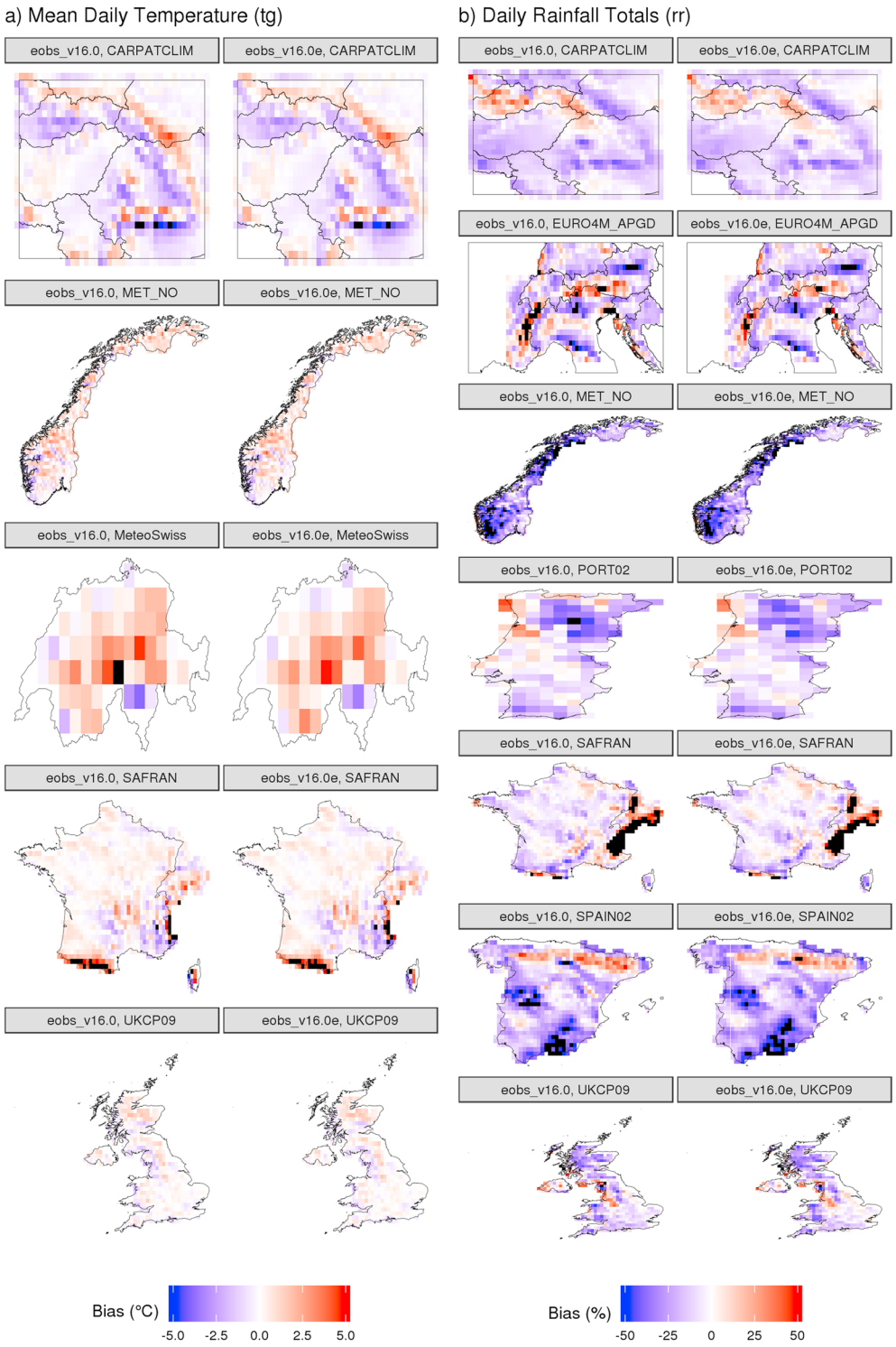



**Figure 9.** Annual climatological differences between E-OBS v16.0e and v16.0 compared to a selection of gridded data sets produced by NMS across Europe for tg (a) and rr (b). The annual averages are calculated over the period 1971–2010 and then the differences derived. In the case of precipitation the differences are expressed as a proportion of the NMS average totals. Grid box values beyond the respective color ranges are marked in black. NMS = National Meteorological Service.



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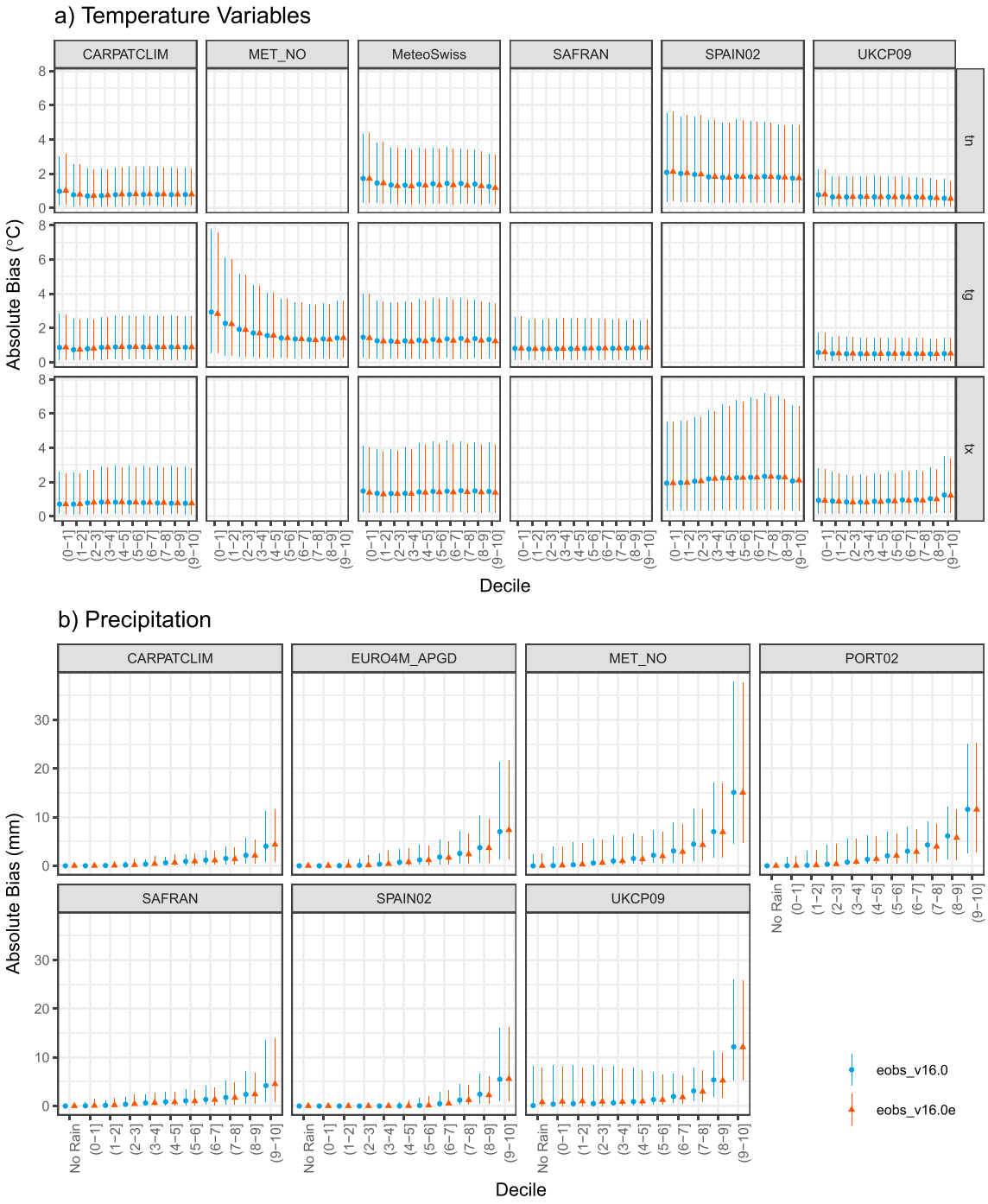



**Figure 10.** Absolute bias (E-OBS *minus* NMS data) in tg, tn, and tx (a) and rr (b) calculated from daily values over the period 1971–2010 grouped by decile determined by the NMS gridded data. The points indicate the median in each category, and the lines indicate the 10th–90th percentile. NMS = National Meteorological Service.

interpolation was used. No correction was applied for elevation differences between E-OBS, and the NMSs data, since the main reason for conducting this comparison was to evaluate the two versions of E-OBS relative to the NMSs data.

In Figure 9 we have plotted the differences (E-OBS *minus* NMSs) in annual climatologies between the data sets for the variables tg and rr. These have been calculated over the period 1971–2010, which is common to all data sets. In general, the differences in the two versions of E-OBS are comparable; the results for tx and tn are similar to those shown here for tg. In the case of tg, the smallest differences occur relative to the UKCP09 data set,



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with the largest differences apparent when compared against the MeteoSwiss tg data set, where both E-OBS data sets are warmer on average by more than 5 ° C across the high Alps. This is likely the result of there being more high-elevation stations available for the MeteoSwiss interpolation. In the case of precipitation, both E-OBS versions show a dry bias relative to the NMSs data sets across most lower elevation regions, although a wet bias is apparent across most mountainous regions. There is an indication that this bias is substantially reduced in the new version of E-OBS when compared against the CARPATCLIM, EURO4M_APGD PORT02, and SPAIN02 data sets.

To examine differences in daily extremes between the two versions of E-OBS, relative to the NMSs gridded data sets, we have sorted the absolute differences in daily values at each grid box into deciles (Figure 10). The deciles are determined by the NMSs grid box values. For precipitation the deciles are calculated for nonzero values, but a further category for zero precipitation values is included. Hofstra et al. (2008) performed a similar comparison in their examination of spatial interpolation methods, as a first stage in developing the original version of E-OBS.

The results indicate little difference in bias between the two versions of E-OBS, relative to the NMSs data. The differences between E-OBS v16.0 and v16.0e differ in only one case: for the variable tx, slightly higher biases in v16.0e are evident relative to the MeteoSwiss gridded data. In the case of precipitation, both of the data sets display the expected scaling of error with interpolated precipitation totals. In general, the results from both this analysis and the comparison of climatological biases indicate that the difference in bias between E-OBS versions are generally small compared to the actual differences between respective versions of E-OBS and the NMSs gridded data. This corresponds to the general finding of Hofstra et al. (2009), who evaluated gridding techniques for the ECA&D stations data relative to station data and NMSs gridded data sets in the development of the old version of E-OBS. Their study concluded that the interpolation technique had less of an effect on the quality of the gridded data compared to the density and quality of input station data. While station density governs the biases observed here, the different interpolation methods and DEM data used in the NMSs gridded data sets probably also have an effect on the results.

## 4. Conclusions

In this paper we have described the construction of a new version of the E-OBS temperature (tn, tg, and tx) and precipitation data sets. A 100-member ensemble of each daily field has been generated, which provides an improved estimation of uncertainty. In terms of both climatological averages and extreme values, the ensemble-mean grids in the new version of E-OBS are broadly comparable to the best guess grids in the earlier version and we stress that station coverage is the most important factor in determining the success of the gridded data. Nonetheless, users will notice differences between the two E-OBS versions, and this results from the different gridding methods used in the two versions of the data set. Furthermore, while the ensemble mean can be taken as grid box averages, the individual ensemble members display a spatial variation that is between a point and a box average value, although this discrepancy varies across the domain in relation to station density.

The uncertainty quantified by the ensemble data set is more closely related to station density than uncertainty values in the original data set, although the uncertainty in data-sparse regions, while much larger in these areas than in the original data set, still appears to be an underestimate of the true uncertainty. The uncertainty quantified by the ensemble only relates to interpolation uncertainty, and improvements may be made in this estimate through the quantification of other sources of uncertainty such as instrumentation error (e.g., Yang et al., 1999b). More general improvements may be made to the E-OBS data set through the use of additional explanatory variables such as slope/aspect, in the case of precipitation, or coastal proximity for temperature (Daly et al., 2008; Vose et al., 2014), or the embedding of non-Gaussian distributions such as the Tweedie distributions (Hasan & Dunn, 2011) in the spatial trend model. Changes to the relative scaling of altitude to longitude/latitude coordinates could also be investigated in further updates to the data set.

In this paper, the ensemble data set has been produced to match the 0.25 ° grid resolution of earlier versions of E-OBS (regular grid spacing). A need exists to produce a version of E-OBS at a higher spatial resolution than the current approximately 25-km resolution (Moreno & Hasenauer, 2016). A higher-resolution version of E-OBS is planned. In addition, the original E-OBS gridding scheme has been used to construct a data set of temperature and precipitation for Southeast Asia (van den Besselaar et al., 2017), and the methods demonstrated in this paper will also be tested for that region, with a view to developing an ensemble data set for the Southeast Asia

region. Furthermore, it remains the case that many of the input station series have not been homogenized and at present we caution against the use of E-OBS for evaluating trends. Efforts are currently underway to produce a version of E-OBS using homogenized station data.

**Acknowledgments**
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